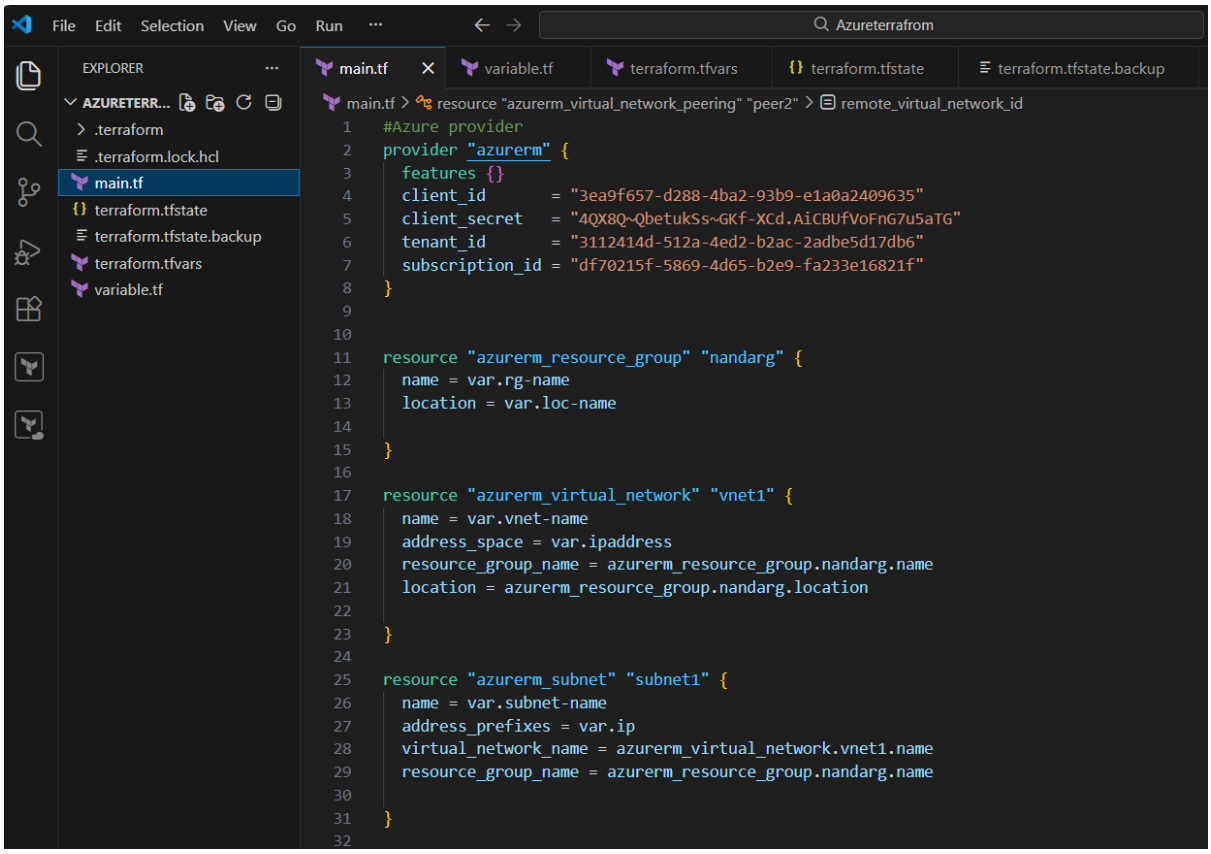
Peering Using Terraform (variables)

- Creating resource Group, virtual Networks, Subnets, Peering in the main.tf file.
- main.tf file for resource definition.
- Calling the values from the terraform.tfvars.
- By terraform init – it will install the terraform plugins.
- terraform.tfstate it store updates



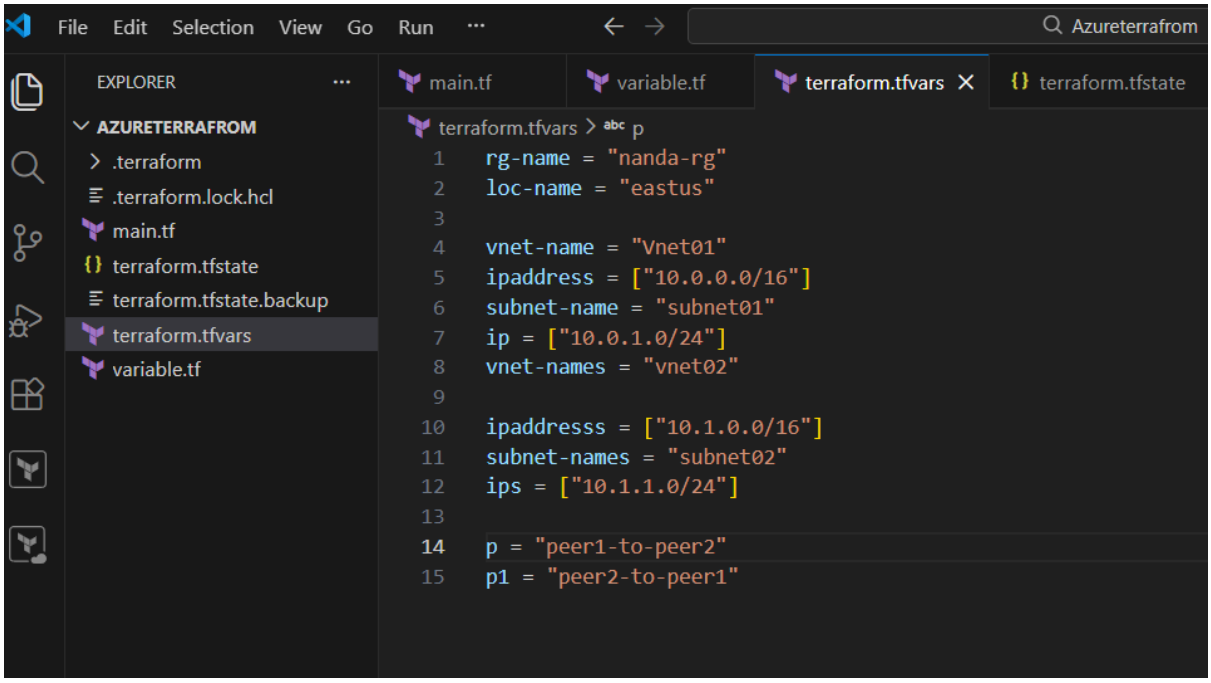
- ✓ In virtual network resource_group_name already initialized so we have to just refer from above.
- ✓ Resource-group-name-- "Azurerm_resource_group.reference_name.name"
- ✓ Location – "Azurerm_resource_group.reference_name.location"

```tf
resource "azurerm_virtual_network" "vnet2" {
  name = var.vnet-names
  address_space = var.ipaddresss
  resource_group_name = azurerm_resource_group.nandarg.name
  location = azurerm_resource_group.nandarg.location

}

resource "azurerm_subnet" "subnet2" {
  name = var.subnet-names
  address_prefixes = var.ips
  virtual_network_name = azurerm_virtual_network.vnet2.name
  resource_group_name = azurerm_resource_group.nandarg.name

}

resource "azurerm_virtual_network_peering" "peer1" {
  name = var.p
  resource_group_name = azurerm_resource_group.nandarg.name
  virtual_network_name = azurerm_virtual_network.vnet2.name
  remote_virtual_network_id = azurerm_virtual_network.vnet1.id
}

resource "azurerm_virtual_network_peering" "peer2" {
  name = var.p1
  resource_group_name = azurerm_resource_group.nandarg.name
  virtual_network_name = azurerm_virtual_network.vnet1.name
  remote_virtual_network_id = azurerm_virtual_network.vnet2.id
}
```

➢ Declaring the varaibles in the variable.tf file

```tf
variable "rg-name"{}

variable "loc-name" {}

variable "vnet-name" {}

variable "ipaddress" {}

variable "subnet-name" {}

variable "ip" {}

variable "vnet-names" {}

variable "ipaddresss" {}

variable "subnet-names" {}

variable "ips" {}

variable "p" {}

variable "p1" {}
```

- Initializing the values in Separate file ---terraform.tfvars.
- Giving values for the variables, which is defined in the variable.tf.



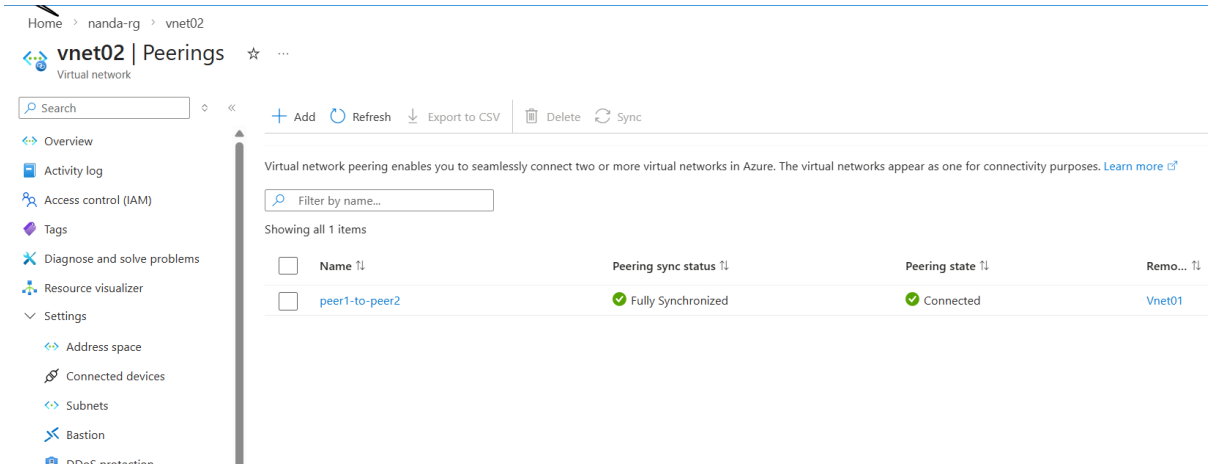
- Terraform **init** -- Initializes Provider.
- Terraform **validate** -- Configuration.
- Terraform **plan** – Shows what will be Created.
- Terraform **apply** – Create Resources.
- Terraform **destroy** – delete.

```
PS C:\Azureterrafrom> terraform apply --auto-approve
azurerm_resource_group.nandarg: Refreshing state... [id=/subscriptions/df70215f-5869-4d65-b2e9-fa23
azurerm_virtual_network.vnet2: Refreshing state... [id=/subscriptions/df70215f-5869-4d65-b2e9-fa233
rk/virtualNetworks/vnet02]
azurerm_virtual_network.vnet1: Refreshing state... [id=/subscriptions/df70215f-5869-4d65-b2e9-fa233
rk/virtualNetworks/Vnet01]
azurerm_subnet.subnet1: Refreshing state... [id=/subscriptions/df70215f-5869-4d65-b2e9-fa233e16821f
ualNetworks/Vnet01/subnets/subnet01]

Terraform used the selected providers to generate the following execution plan. Resource actions ar
following symbols:
  + create

Terraform will perform the following actions:

  # azurerm_subnet.subnet2 will be created
  + resource "azurerm_subnet" "subnet2" {
      + address_prefixes                            = [
          + "10.1.1.0/24",
        ]
      + default_outbound_access_enabled             = true
      + id                                          = (known after apply)
      + name                                        = "subnet02"
      + private_endpoint_network_policies           = "Disabled"
      + private_link_service_network_policies_enabled = true
      + resource_group_name                         = "nanda-rg"
      + virtual_network_name                        = "vnet02"
    }

Plan: 1 to add, 0 to change, 0 to destroy.
azurerm_subnet.subnet2: Creating...
azurerm_subnet.subnet2: Creation complete after 8s [id=/subscriptions/df70215f-5869-4d65-b2e9-fa233
rk/virtualNetworks/vnet02/subnets/subnet02]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.
```

```
PS C:\Azureterrafrom> terraform apply --auto-approve
azurerm_resource_group.nandarg: Refreshing state... [id=/subscriptions/df70215f-5869-4d65-b2e9-fa233e16821f/
azurerm_virtual_network.vnet2: Refreshing state... [id=/subscriptions/df70215f-5869-4d65-b2e9-fa233e16821f/
rk/virtualNetworks/vnet02]
azurerm_virtual_network.vnet1: Refreshing state... [id=/subscriptions/df70215f-5869-4d65-b2e9-fa233e16821f/
rk/virtualNetworks/Vnet01]
azurerm_subnet.subnet2: Refreshing state... [id=/subscriptions/df70215f-5869-4d65-b2e9-fa233e16821f/resourc
ualNetworks/vnet02/subnets/subnet02]
azurerm_subnet.subnet1: Refreshing state... [id=/subscriptions/df70215f-5869-4d65-b2e9-fa233e16821f/resourc
ualNetworks/Vnet01/subnets/subnet01]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicat
following symbols:
  + create

Terraform will perform the following actions:

  # azurerm_virtual_network_peering.peer1 will be created
  + resource "azurerm_virtual_network_peering" "peer1" {
      + allow_forwarded_traffic                = false
      + allow_gateway_transit                  = false
      + allow_virtual_network_access           = true
      + id                                     = (known after apply)
      + name                                   = "peer1-to-peer2"
      + peer_complete_virtual_networks_enabled = true
      + remote_virtual_network_id              = "/subscriptions/df70215f-5869-4d65-b2e9-fa233e16821f/resour
rtualNetworks/Vnet01"
      + resource_group_name                    = "nanda-rg"
      + use_remote_gateways                    = false
      + virtual_network_name                   = "vnet02"
    }

  # azurerm_virtual_network_peering.peer2 will be created
  + resource "azurerm_virtual_network_peering" "peer2" {
      + allow_forwarded_traffic                = false
      + allow_gateway_transit                  = false
      + allow_virtual_network_access           = true
      + id                                     = (known after apply)
      + name                                   = "peer2-to-peer1"
      + peer_complete_virtual_networks_enabled = true
      + remote_virtual_network_id              = "/subscriptions/df70215f-5869-4d65-b2e9-fa233e16821f/resour
rtualNetworks/vnet02"
      + resource_group_name                    = "nanda-rg"
      + use_remote_gateways                    = false
      + virtual_network_name                   = "Vnet01"
    }
```

```
    }

Plan: 2 to add, 0 to change, 0 to destroy.
azurerm_virtual_network_peering.peer2: Creating...
azurerm_virtual_network_peering.peer1: Creating...
azurerm_virtual_network_peering.peer2: Still creating... [00m10s elapsed]
azurerm_virtual_network_peering.peer1: Still creating... [00m10s elapsed]
azurerm_virtual_network_peering.peer1: Still creating... [00m20s elapsed]
azurerm_virtual_network_peering.peer2: Still creating... [00m20s elapsed]
azurerm_virtual_network_peering.peer2: Creation complete after 30s [id=/subscripti
/Microsoft.Network/virtualNetworks/Vnet01/virtualNetworkPeerings/peer2-to-peer1]
azurerm_virtual_network_peering.peer1: Still creating... [00m30s elapsed]
azurerm_virtual_network_peering.peer1: Still creating... [00m40s elapsed]
azurerm_virtual_network_peering.peer1: Still creating... [00m50s elapsed]
azurerm_virtual_network_peering.peer1: Creation complete after 58s [id=/subscripti
/Microsoft.Network/virtualNetworks/vnet02/virtualNetworkPeerings/peer1-to-peer2]

Apply complete! Resources: 2 added, 0 changed, 0 destroyed.
PS C:\Azureterrafrom>
```

- Peering is Done.



➢ Terraform destroy is to delete.

```
PS C:\Azureterrafrom> terraform  destroy
azurerm_resource_group.nandarg: Refreshing state... [id=/subscriptions/df70215f-5869-4d65-b2e9-fa233e16821
azurerm_virtual_network.vnet1: Refreshing state... [id=/subscriptions/df70215f-5869-4d65-b2e9-fa233e16821-
rk/virtualNetworks/Vnet01]
azurerm_virtual_network.vnet2: Refreshing state... [id=/subscriptions/df70215f-5869-4d65-b2e9-fa233e16821-
rk/virtualNetworks/vnet02]
azurerm_subnet.subnet1: Refreshing state... [id=/subscriptions/df70215f-5869-4d65-b2e9-fa233e16821f/resou
ualNetworks/Vnet01/subnets/subnet01]
azurerm_subnet.subnet2: Refreshing state... [id=/subscriptions/df70215f-5869-4d65-b2e9-fa233e16821f/resou
ualNetworks/vnet02/subnets/subnet02]
azurerm_virtual_network_peering.peer2: Refreshing state... [id=/subscriptions/df70215f-5869-4d65-b2e9-fa2
ft.Network/virtualNetworks/Vnet01/virtualNetworkPeerings/peer2-to-peer1]
azurerm_virtual_network_peering.peer1: Refreshing state... [id=/subscriptions/df70215f-5869-4d65-b2e9-fa2
ft.Network/virtualNetworks/vnet02/virtualNetworkPeerings/peer1-to-peer2]

Terraform used the selected providers to generate the following execution plan. Resource actions are indi
  - destroy

Terraform will perform the following actions:

  # azurerm_resource_group.nandarg will be destroyed
  - resource "azurerm_resource_group" "nandarg" {
      - id         = "/subscriptions/df70215f-5869-4d65-b2e9-fa233e16821f/resourceGroups/nanda-rg" -> nul
      - location   = "eastus" -> null
      - name       = "nanda-rg" -> null
      - tags       = {} -> null
        # (1 unchanged attribute hidden)
    }

  # azurerm_subnet.subnet1 will be destroyed
  - resource "azurerm_subnet" "subnet1" {
      - address_prefixes                            = [
          - "10.0.1.0/24",
        ] -> null
      - default_outbound_access_enabled             = true -> null
      - id                                          = "/subscriptions/df70215f-5869-4d65-b2e9-fa233e168
work/virtualNetworks/Vnet01/subnets/subnet01" -> null
      - name                                        = "subnet01" -> null
      - private_endpoint_network_policies           = "Disabled" -> null
```

```
Plan: 0 to add, 0 to change, 7 to destroy.

Do you really want to destroy all resources?
  Terraform will destroy all your managed infrastructure, as shown above.
  There is no undo. Only 'yes' will be accepted to confirm.

  Enter a value: yes

azurerm_subnet.subnet2: Destroying... [id=/subscriptions/df70215f-5869-4d65-b2e9-fa233e16821f/resourceGroups/nanda
works/vnet02/subnets/subnet02]
azurerm_subnet.subnet1: Destroying... [id=/subscriptions/df70215f-5869-4d65-b2e9-fa233e16821f/resourceGroups/nanda
works/Vnet01/subnets/subnet01]
azurerm_virtual_network_peering.peer2: Destroying... [id=/subscriptions/df70215f-5869-4d65-b2e9-fa233e16821f/resou
work/virtualNetworks/Vnet01/virtualNetworkPeerings/peer2-to-peer1]
azurerm_virtual_network_peering.peer1: Destroying... [id=/subscriptions/df70215f-5869-4d65-b2e9-fa233e16821f/resou
work/virtualNetworks/vnet02/virtualNetworkPeerings/peer1-to-peer2]
azurerm_subnet.subnet2: Still destroying... [id=/subscriptions/df70215f-5869-4d65-b2e9-...irtualNetworks/vnet02/su
azurerm_subnet.subnet1: Still destroying... [id=/subscriptions/df70215f-5869-4d65-b2e9-...irtualNetworks/Vnet01/su
azurerm_virtual_network_peering.peer2: Still destroying... [id=/subscriptions/df70215f-5869-4d65-b2e9-.../virtualN
sed]
azurerm_virtual_network_peering.peer1: Still destroying... [id=/subscriptions/df70215f-5869-4d65-b2e9-.../virtualN
sed]
azurerm_subnet.subnet1: Destruction complete after 12s
azurerm_virtual_network_peering.peer2: Destruction complete after 12s
azurerm_subnet.subnet2: Destruction complete after 12s
azurerm_virtual_network_peering.peer1: Still destroying... [id=/subscriptions/df70215f-5869-4d65-b2e9-.../virtualN
sed]
azurerm_virtual_network_peering.peer1: Destruction complete after 24s
azurerm_virtual_network.vnet2: Destroying... [id=/subscriptions/df70215f-5869-4d65-b2e9-fa233e16821f/resourceGroup
tualNetworks/vnet02]
azurerm_virtual_network.vnet1: Destroying... [id=/subscriptions/df70215f-5869-4d65-b2e9-fa233e16821f/resourceGroup
tualNetworks/Vnet01]
azurerm_virtual_network.vnet2: Still destroying... [id=/subscriptions/df70215f-5869-4d65-b2e9-...crosoft.Network/v
azurerm_virtual_network.vnet1: Still destroying... [id=/subscriptions/df70215f-5869-4d65-b2e9-...crosoft.Network/v
azurerm_virtual_network.vnet2: Destruction complete after 12s
azurerm_virtual_network.vnet1: Destruction complete after 12s
azurerm_resource_group.nandarg: Destroying... [id=/subscriptions/df70215f-5869-4d65-b2e9-fa233e16821f/resourceGrou
azurerm_resource_group.nandarg: Still destroying... [id=/subscriptions/df70215f-5869-4d65-b2e9-fa233e16821f/resour
azurerm_resource_group.nandarg: Still destroying... [id=/subscriptions/df70215f-5869-4d65-b2e9-fa233e16821f/resour
azurerm_resource_group.nandarg: Destruction complete after 24s

Destroy complete! Resources: 7 destroyed.
PS C:\Azureterrafrom>
```